

Prospective Life Cycle Assessment and Input-Output Modeling of Polylactic Acid Production: A 2050 Sustainability Roadmap for Denmark

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1. Introduction

1.1 Project Overview and Case Specifics

This report evaluates the environmental performance of Polylactic Acid (PLA), a bio-based and biodegradable alternative to conventional fossil-based plastics. Based on Case No. 48, the study focuses on the transformation phase of PLA granulate derived from renewable agricultural feedstocks such as corn and sugar crops. The analysis adopts a cradle-to-gate boundary, including upstream raw material extraction and international logistics.

1.2 Strategic Importance of Prospective Modeling for PLA

Evaluating the future sustainability of PLA is important because its environmental performance depends on both energy-system decarbonization and agricultural feedstock production. Dynamic modeling is therefore needed to assess burden-shifting risks and the potential to achieve sustainability targets by 2050. To support this assessment, SSP1 and SSP2 scenarios were applied to represent different levels of climate-policy ambition and technological development.

1.3 Macro-sector Definition and Mapping

To contextualize PLA production within the global economy, the process was mapped to the EXIOBASE 3 sector “Plastics, basic” (Sector 24.3), representing primary polymer production and related background activities.

Sectoral Characteristics and Scale in Denmark: In 2019, the sector accounted for 0.78% of Denmark’s total regional output, with a total output of 4,470.12 MEUR.

Supply-Demand Dynamics and Z-Matrix Insights: By examining the Inter-industry Requirements (Z-matrix) and Final Demand (Y-matrix), the study evaluates the region's self-sufficiency. Denmark exhibits a Self-sufficiency Ratio (SSR) of 90.78% for basic plastics. While a significant portion of production is domestic, the remaining 9.22% (representing a demand gap of approximately 373 MEUR) is fulfilled through international imports. This structural reliance on the global market explains the sector's integration with international logistics and foreign biochemical supply chains. Consequently, the environmental performance of PLA transformed in Denmark is inherently tied to the decarbonization of both domestic energy grids and global transport infrastructures.

Baseline Impact Intensity and Hotspots: The benchmark greenhouse gas (GHG) impact intensity for this sector was determined to be 0.286 kg CO_2e/EUR. Structural Path Analysis (SPA) further confirms that environmental hotspots are not limited to direct factory emissions but are significantly concentrated in Step 2 (First-tier suppliers), specifically within maritime logistics and thermal energy supply services required for feedstock processing.

Environmental Learning Effects and Future Projections: Future environmental performance was evaluated by combining SSP-based background developments with foreground learning effects. While background decarbonization dominates long-term reductions, stronger environmental learning further accelerates improvements over time.

2. Methods

2.1 Life Cycle Assessment (LCA) Approach

2.1.1 Functional Unit (FU) and Reference Flow (RF)

Functional Unit: The functional unit (FU) is defined as the production of 1 kg of industrial grade polylactic acid (PLA) granulate. A mass-based functional unit was selected to ensure consistency with the process data

and impact assessment results generated in Activity Browser, and to provide a clear and comparable basis for evaluating environmental impacts.

Reference Flow: The specific mass and energy flows required to achieve the FU, including lactic acid monomer inputs, electricity and heat consumption during polymerization, and necessary auxiliary chemicals.

2.1.2 Temporal Boundary Statement

The assessment covers two distinct temporal periods to capture dynamic sustainability trends:

Baseline (2019): A static assessment utilizing the most recent empirical data from the EXIOBASE 3 database.

Prospective Target (2050): A future projection accounting for grid decarbonization, technological learning effects (eLRs), and socio-economic changes under specific SSP scenarios.

2.1.3 System Boundary Illustration

The system boundary distinguishes between the foreground (process-based) and background (EEIO-based) systems, reflecting differences in modelling approach and level of detail along the supply chain.

Foreground System (Direct Inputs): Represents the core transformation processes within the Danish plastics sector, specifically the polymerization activities detailed in Case No. 48.

Background System (Indirect Suppliers): Encompasses all other upstream suppliers, modeled through EXIOBASE 3, including agricultural cultivation (e.g., corn/sugar), international maritime logistics, and global energy infrastructures.

Analytical Integration: Structural Path Analysis (SPA) is applied to trace how environmental interventions from the background system (indirect suppliers) accumulate and manifest within the foreground system (direct production).

2.1.4 Region of the Study

The foreground production is geographically defined in Denmark (DK). However, the system depends on imported intermediates and global supply chains, implying that environmental impacts are influenced by both domestic and international processes.

2.1.5 Macro-sector Specification

The foreground system of this study is centered on the EXIOBASE sector *"Plastics, basic"* (Sector 24.3), selected as a proxy for PLA production. Based on this core sector, a macro-sector is defined within the EXIOBASE 3 framework using a customized Python-based filtering algorithm to ensure the empirical integrity of the baseline model.

Algorithmic Definition and Data Cleaning: The selection process employed keyword-based inclusion and categorical exclusion to isolate the primary production phase. Specifically, the script standardized sector names to lowercase and matched the keyword *"plastics, basic"*, while excluding irrelevant categories such as *"secondary"*, *"treatment"*, *"waste"*, *"incineration"*, and *"landfill"*.

Logical Exclusions: To maintain a strict cradle-to-gate scope, the algorithm systematically excluded downstream waste-management and secondary-processing activities. While upstream processes (e.g., agriculture, chemicals, and energy supply) were not explicitly selected, they are implicitly captured within the EEIO framework through inter-industry linkages, representing sectoral-average rather than PLA-specific supply chains.

Methodological Justification: This filtering approach ensures that the system focuses on primary polymer production while excluding downstream processes, maintaining consistency with the defined system boundary while balancing representativeness and data availability within the EEIO framework.

2.2 Prospective Approach

This study evaluates the environmental profile of Polylactic Acid (PLA) production by integrating the macro-evolution of background systems with micro-level technological learning in the foreground system. The overall methodological workflow is illustrated in the flowchart below:

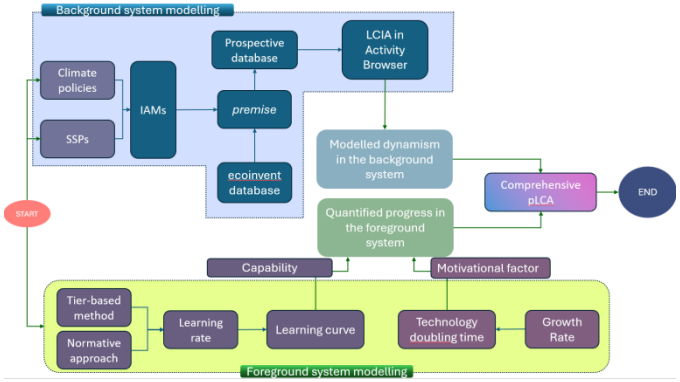


Figure 1. Workflow of the prospective LCA modeling for PLA production.

Methodological Framework and Tools: To simulate environmental impacts from 2025 to 2050, this study uses Activity Browser together with the premise framework to transform the ecoinvent 3.0 database into future scenario-based background systems. Premise links IAM outputs with LCA databases by modifying parameters such as energy mix, technology efficiency, and emission intensities. Due to the reduced-size database, global (GLO) datasets were used where regional data were unavailable, as the study focuses on relative trends rather than absolute values.

Background System (IAM Scenarios and SSPs): Prospective background systems were modeled using the REMIND IAM under SSP1 and SSP2 for 2020–2050. SSP1 represents rapid decarbonization and ambitious climate policies, while SSP2 serves as a moderate reference pathway. Comparing these scenarios allows assessment of how different background developments influence PLA environmental performance, particularly for electricity, heat, fuels, and transport systems.

Foreground System (Environmental Learning Curves): While the background system evolves via IAMs, PLA foreground production was modeled using an S-curve learning model to represent different stages of technological development. The environmental learning rate (eLR) was derived from the relationship between cumulative production and environmental impact reduction.

The annual growth rate of cumulative production was calculated assuming exponential growth:

$$S_t = S_0(1+r)^{\Delta t}$$

Where S_0 and S_t represent the cumulative production (or installed capacity) at the initial and final time points, respectively, and Δt is the time interval between them.

The corresponding technology doubling time T_d, representing the time required for cumulative production to double, was calculated as:

$$T_d = t\frac{\ln(2)}{\ln(1+r)}$$

Finally, eLR defined as the percentage reduction in environmental impact per doubling of cumulative production, was expressed as:

$$eLR = (1-2^{-\alpha_e}) \times 100\%$$

Where α_e is the environmental learning exponent describing the sensitivity of environmental impact to experience accumulation.

In addition to forward projections, a backcasting approach was applied to estimate the hypothetical learning

rate (hLR) required to achieve predefined environmental targets. Detailed formulations are provided in Appendix III.

2.3 Life Cycle Inventory (LCI)

The LCI was constructed in Activity Browser using a representative PLA production process from the ecoinvent database, consistent with the defined functional unit of 1 kg PLA granulate. The global dataset "polylactic acid granulate {GLO} | production | cut-off, U" was selected, reflecting the globally distributed nature of PLA supply chains.

The selected process represents the production stage of PLA and includes relevant upstream activities (e.g., raw materials, energy, and transport), ensuring a consistent cradle-to-gate system boundary. The background system was updated through scenario-based relinking using the premise tool, applying SSP1 and SSP2 scenarios based on the REMIND model to capture future changes in energy systems.

2.4 Life Cycle Impact Assessment (LCIA)

The Life Cycle Impact Assessment (LCIA) was performed using the Environmental Footprint method (EF v3.0), which provides a harmonized set of midpoint impact indicators for assessing environmental sustainability across multiple impact categories. The method is widely used in European policy contexts and ensures consistency in impact characterization.

In this study, three impact categories were selected for analysis:

- Climate change (GWP100, kg CO_2-eq)
- Ozone depletion (ODP, kg CFC-11-eq)
- Material resources (ADP, kg Sb-eq)

These indicators were chosen to capture different environmental mechanisms and to identify potential trade-offs across impact categories. The LCIA results were calculated for each scenario (SSP1 and SSP2) and time step (2020–2050). The analysis focuses on comparing relative trends across scenarios rather than absolute values, allowing for the assessment of how different background system developments influence environmental impacts over time.

While the EF method provides comprehensive coverage of midpoint environmental impacts, it does not directly represent endpoint damages or absolute sustainability thresholds. In addition, the characterization factors are based on generalized assumptions and may not fully capture regional or technology-specific variations

3. Results

3.1 Life Cycle Impact Assessment Results

3.1.1 Production and Consumption-based Impacts of the Macro-sector

At a global scale, EXIOBASE 3 calculations show balanced production-based (PBA) and consumption-based (CBA) impacts at approximately 49.75 Gt CO_2e. However, the Danish basic plastics sector exhibits a significant divergence: a PBA of 10.83 Mt CO_2e versus a CBA of 17.73 Mt CO_2e. This discrepancy identifies Denmark as a net importer of embodied carbon, reflecting the heavy reliance of high-income consumption patterns on global supply chains. Although the sector represents only 0.03% of global emissions, its consumption-driven environmental pressure remains substantial.

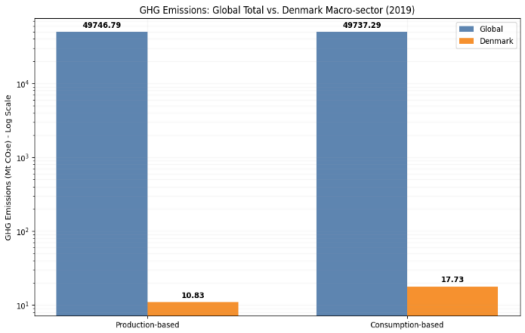


Figure 2. PBA vs. CBA GHG Emissions: Global and Denmark Basic Plastics Sector Comparison.

3.1.2 Significant Environmental Trends and Hotspot Analysis

Quantitative modeling identified a GHG emission intensity of 0.2864 kg CO_2e/EUR for the Danish plastics sector. Structural Path Analysis (SPA) showed that the largest contributors were domestic maritime transport (0.056 kt CO_2e/MEUR), direct sectoral emissions, and steam supply, highlighting strong dependence on fossil-fuel-based logistics and energy systems.

Tier decomposition further revealed that first-tier suppliers (Step 2) represent the main environmental hotspot, contributing approximately 0.087 kt CO_2e/MEUR. Key pathways include thermal energy supply and international maritime transport. The results indicate a burden-shifting risk, where environmental pressures increasingly move from the foreground system toward background supply chains such as international shipping.

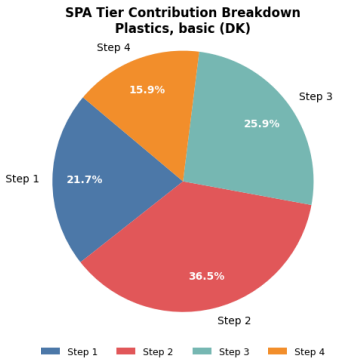


Figure 3. SPA Tier Contribution of the Danish Basic Plastics Macro-sector (2019).

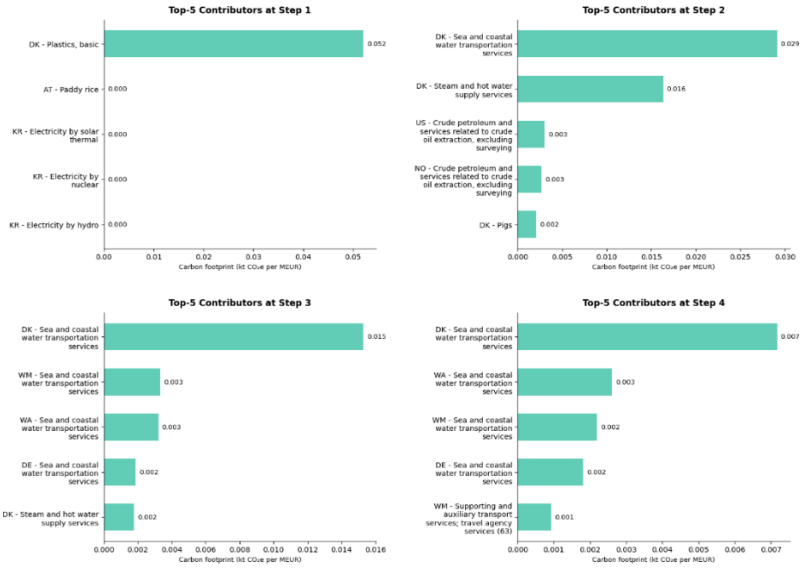


Figure 4.Top 5 Contributing Sectors Across the First Four Supply Chain Tiers (Step 1 to Step 4) for DK Plastic Industry.

3.1.3 Projected Environmental Impacts (2020–2050)

The environmental impacts were evaluated using a representative PLA process from the ecoinvent database (GLO). The analysis focuses on relative trends under SSP scenarios.

Climate change (CO_2): Climate change impacts decrease over time in both SSP1 and SSP2, with a stronger reduction under SSP1 due to faster decarbonization. Specifically, impacts decline from 2.63 to 0.79 kg CO_2-eq/FU under SSP1 and from 2.67 to 1.41 kg CO_2-eq/FU under SSP2 between 2020 and 2050.

Ozone depletion (ODP): In contrast, ozone depletion increases over time under both scenarios, suggesting additional emissions associated with technological and infrastructure changes. Values increase from 2.91E-08

to 3.67E-08 kg CFC-11-eq/FU under SSP1 and from 2.91E-08 to 3.64E-08 kg CFC-11-eq/FU under SSP2.

Material resources: Material resource use also increases over time, reflecting growing demand for metals and minerals in low-carbon systems. The impact rises from 1.36E-05 to 2.05E-05 kg Sb-eq/FU under SSP1 and from 1.36E-05 to 1.80E-05 kg Sb-eq/FU under SSP2.

Overall, the results reveal contrasting trends across impact categories. While climate change impacts decrease due to energy system decarbonization, ozone depletion and material resource use increase, suggesting potential burden shifting. This indicates that future low-carbon pathways may reduce CO_2 emissions but increase pressures related to resource extraction and infrastructure development.

SPA results further help explain these patterns. Upstream processes, particularly maritime transport and thermal energy supply, are key contributors to the environmental profile of PLA production. As decarbonization progresses, environmental burdens shift from direct fossil emissions toward upstream supply chains and infrastructure-related processes. This highlights the importance of considering multiple impact categories when assessing the future environmental performance of PLA.

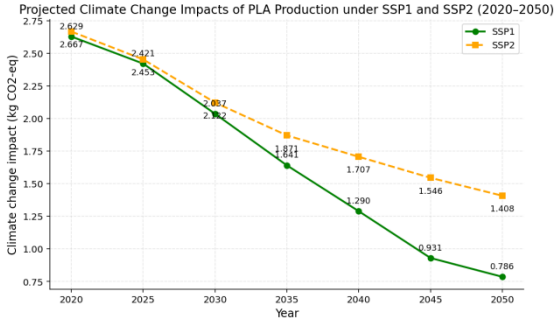


Figure 5. Projected climate change impacts of PLA production under SSP1 and SSP2 (2020–2050).

3.2 Background System Analysis

Influence of background system changes: The background system strongly influences PLA environmental performance through upstream processes such as electricity, fuels, transport, and material production. As shown in Section 3.1.3, climate change impacts decrease over time, with larger reductions under SSP1 due to faster energy-system decarbonization and renewable electricity deployment. SSP2 follows a slower transition, resulting in higher residual emissions.

Trade-offs in development trajectories: Ozone depletion and material resource use increase under both scenarios, indicating trade-offs where CO_2 reductions are partly offset by higher impacts in other categories. Expansion of low-carbon infrastructure increases demand for metals, minerals, and upstream industrial activities, shifting environmental burdens rather than fully eliminating them.

Electricity system (background record analysis): Contribution analysis results (Appendix II) show that under SSP1, CO_2 contributions decrease significantly over time due to electricity-system decarbonization, while methane, sulfur dioxide, and urban air pollutants become relatively more important. In contrast, SSP2 retains a larger CO_2 contribution because of continued fossil-fuel dependence, explaining the smaller climate-impact reductions observed in Section 3.1.3.

Climate policies and societal behaviors: SSP1 assumes faster renewable-energy deployment and stronger climate policies, while SSP2 represents a slower transition pathway. Increasing demand for bio-based products may further support PLA market growth and technological learning. Comparing SSP1 and SSP2 highlights important trade-offs and hotspots, particularly in maritime transport, energy infrastructure, and resource extraction.

3.3 Environmental Learning Effects and Motivational Factors

3.3.1 Environmental learning-based projection of foreground impacts

Environmental learning effects describe the reduction of environmental impacts with increasing cumulative production. In this study, a quasi S-curve model is used to represent the non-linear evolution of PLA technology. The model is parameterized using a plateau factor (M = 0.4) and a shape parameter (B = 0.3), which control the long-term saturation level and curvature of the learning trajectory. The environmental learning rate (eLR), representing technological capability, is set to 3.3%(default), 5%, 10%, and 15%. The technology doubling time is fixed at 3.56 years based on the 2024 European Bioplastics market update[1], reflecting rapid market expansion.

Figure 6 compares background, foreground, and comprehensive impacts. The background system, based on REMIND IAM scenarios, shows a steady decline driven by decarbonization, while the foreground system decreases more gradually due to accumulated learning. The overall impact reflects their combination, with background changes dominating long-term trends and foreground learning providing additional reductions. These projections are subject to limitations, as REMIND may not capture PLA-specific dynamics and the learning model simplifies technological development.

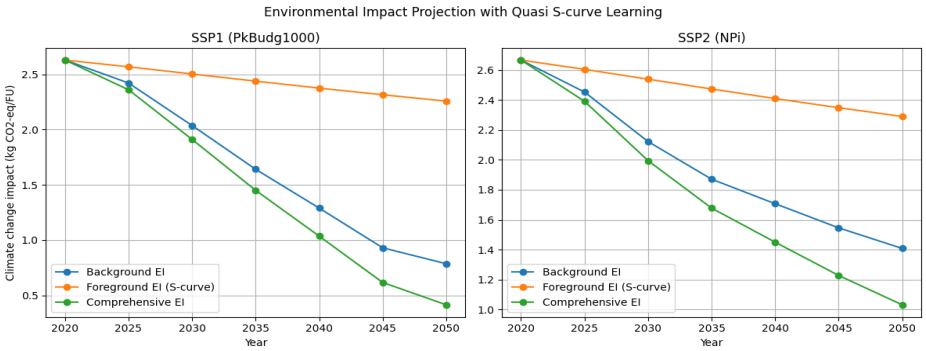


Figure 6. Projected Environmental Impacts under Quasi S-curve Learning for SSP1 and SSP2 Scenarios.

Figure 7 shows that higher eLR values lead to faster reductions in environmental impacts. For example, eLR values of 5–15% result in significantly lower impacts by 2050 compared to 3.3%, indicating that stronger technological capability accelerates environmental improvement, although overall trends remain constrained by the background system.

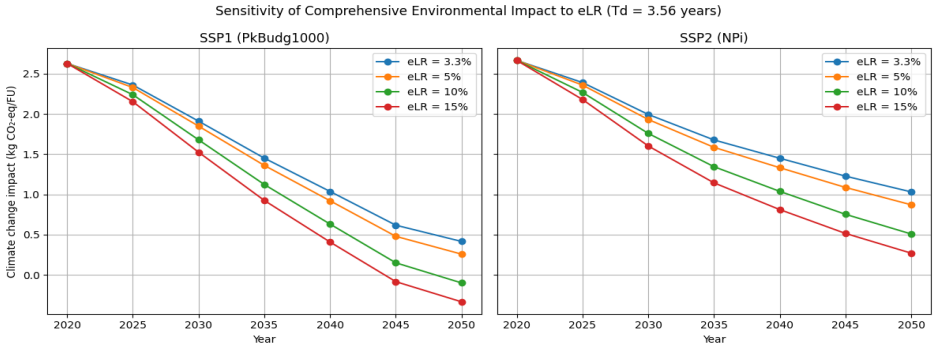


Figure 7. Impact of Environmental Learning Rates on Comprehensive Environmental Performance (T_d = 3.56 years).

3.3.2 Influence of technological motivation (doubling time)

Technological progress depends not only on learning capability but also on technological motivation, represented by the technology doubling time (T_d). According to the 2024 European Bioplastics market update[1], global bioplastics production capacity is projected to increase from 2.47 million tons in 2024 to 5.73 million tons in 2029, while the share of PLA rises from 37.1% to 42.3%. Based on these values, global

PLA capacity increases from approximately 0.916 to 2.424 million tons, corresponding to a compound annual growth rate of 21.5% and a doubling time of 3.56 years. A conservative scenario with T_d = 10 years is also considered to reflect slower market expansion.

Table 1 presents the comprehensive environmental impacts (kg CO_2-eq/FU) under both scenarios at a fixed learning rate (eLR = 5%). The results show a clear divergence over time. In 2035, impacts are 1.36 (T_d = 3.56y) versus 1.54 kg CO_2-eq/FU (T_d = 10y), a difference of about 13%. By 2045, this gap increases to 0.48 versus 0.76 (≈58%), and by 2050 to 0.26 versus 0.59, with the conservative scenario showing more than twice the impact.

Table 1. Comparison of Comprehensive EI (kg CO_2-eq/FU) under Different Doubling Time Scenarios (eLR = 5%).

Year	2020	2025	2030	2035	2040	2045	2050
T_d = 3.56 y	2.63	2.33	1.85	1.36	0.92	0.48	0.26
T_d = 10 y	2.63	2.39	1.97	1.54	1.16	0.76	0.59

These results demonstrate that shorter doubling times accelerate cumulative learning and lead to significantly faster environmental improvements. Under slower expansion, achieving the same environmental targets would require either more time or higher learning rates. This highlights the complementary roles of capability (eLR) and motivation (T_d) in driving environmental progress.

3.3.3 Required learning conditions to achieve environmental targets

To assess whether PLA production can achieve carbon neutrality by 2050, a backcasting approach based on environmental learning theory was applied, with the target defined as 0 kg CO_2-eq/FU for 1 kg PLA. SSP2 was selected as the reference scenario because it represents a middle-of-the-road pathway and avoids overly optimistic background decarbonization.

A Tier II/III literature-based approach identified representative historical learning rates for PLA-related systems. Previous studies reported learning rates of 10–25%[2], with comparable industrial processes around 12.5%[3]. Based on the literature, a representative historical learning rate of 15%, corresponding to a power constant (α) of ~0.23, was selected. Using the Wright-type learning approach with a technology doubling time of Td = 3.56 years, an environmental learning rate (eLR) of 8.52% was estimated, meaning impacts decrease by 8.52% per doubling of cumulative production. When the doubling time increased to 10 years, the required learning rate rose to 22.14%, indicating that slower market growth requires substantially higher technological improvement. The estimated 8.52% remains within the literature range, suggesting that the target is potentially achievable despite uncertainties from proxy rather than PLA-specific historical data. Tier I, IV, and V approaches were excluded because of limited PLA-specific data, higher subjectivity, or insufficient accuracy.

Under SSP1, faster background decarbonization further reduced climate impacts compared with SSP2. Figure 8 shows that background decarbonization alone reduces impacts to 1.41 kg CO_2-eq/FU by 2050 under SSP2, but this remains insufficient for carbon neutrality. Including learning effects enables impacts to approach zero by 2050, although major reductions occur mainly after 2040, indicating that intermediate targets are not achieved in earlier periods.

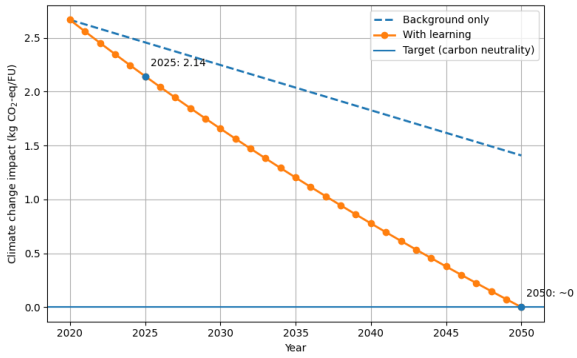


Figure 8. Projected pathway of climate change impact for PLA production and timing of carbon neutrality (2020–2050).

Although climate impacts decrease substantially over time, previous results showed increasing impacts in ozone depletion and material resource use, highlighting the importance of avoiding burden shifting in future PLA development. Additional hLR calculations for climate neutrality and burden-shifting mitigation remained below the literature-based learning-rate range, suggesting that the proposed environmental targets are achievable within the investigated time frame. Under SSP2, the required hLRs were estimated at 4.57% for material resource use and 3.36% for ozone depletion. Detailed results are provided in the Appendix.

3.4 Suggestions for a Sustainable Roadmap

An Absolute Sustainability Assessment (ASA) was conducted to estimate the share of the global Safe Operating Space (SOS) allocated to the Danish PLA system. As direct Danish PLA consumption data were unavailable, consumption was estimated from national plastic use and global bioplastics shares[1,4]. With ~551 kt/year plastic use, 0.5% bioplastics share, and PLA accounting for 37.1% of global capacity in 2024[1], Danish PLA consumption is approximated at ~1 kt/year.

Using per-capita and final consumption expenditure (FCE) allocation, SOS shares of 7.9×10^{-4} and 4.73×10^{-6} were obtained. EF v3.1 results indicate that the Danish PLA system (1.97 kt CO_2-eq/FU) exceeds its allocated climate SOS (2.54×10^{-2} kt CO_2-eq/FU) by 77.6 times, showing that current improvements are insufficient to remain within planetary boundaries.

Under SSP1, prospective learning analysis suggests a required environmental learning rate (eLR) of 10.7% (growth rate 15%, doubling time 4.96 years), indicating that the climate target remains achievable. Sustainable-by-design strategies include renewable energy integration, process optimization, and low-carbon supply chains, with a potential pathway of renewable electricity before 2030, large-scale decarbonization by 2040, and near climate-neutral production by 2050. However, high costs, infrastructure needs, and supply-chain dependence remain key barriers. Supporting tables and figures are provided in the Appendix.

4. Discussion

Macro-sector vs Product Perspective: The comparison between macro-sector and process-level results highlights key differences in impact representation. At the global scale, MRIO results show nearly balanced production- and consumption-based emissions (49.75 Gt CO_2e), whereas the Danish “basic plastics” sector exhibits higher consumption-based emissions (17.73 Mt CO_2e) than production-based emissions (10.83 Mt CO_2e), indicating reliance on imported embodied carbon. In contrast, the PLA process-level assessment shows a clear decline in impacts over time, mainly driven by background system decarbonization and supported by technological improvements. These differences reflect methodological characteristics: EEIO provides system-wide completeness but relies on aggregated data, whereas process-based LCA captures technology-specific dynamics but with limited system boundaries. Therefore, the two approaches are complementary, combining system-level insight with technological detail.

Impact Reduction Strategies: Reducing environmental impacts requires actions at both macro-sector and

product levels. At the macro level, mitigation should focus on reducing embodied emissions in imports, decarbonizing energy systems, and improving low-carbon transport and upstream material production. At the product level, PLA impacts can be reduced through process optimization, higher energy efficiency, and increased use of renewable energy and sustainable feedstocks. These improvements are closely linked to environmental learning effects, where technological progress driven by scaling enables continuous reductions over time.

Burden-Shifting and Trade-offs: The results show that while climate change impacts decrease significantly over time, burden-shifting effects emerge across other impact categories. Under the SSP1 scenario, climate change impacts drop from 2.63 to 0.79 kg CO_2-eq/FU between 2020 and 2050, whereas material resource use increases from 1.36×10^{-5} to 2.05×10^{-5} kg Sb-eq/FU. This indicates clear trade-offs between decarbonization and resource-related impacts.

These trade-offs are mainly driven by structural changes in the background system. While decarbonization of energy supply reduces greenhouse gas emissions, the expansion of low-carbon infrastructure increases demand for metals and minerals, and upstream processes such as transport and heat supply remain key contributors. As a result, environmental pressures shift from direct emissions to the broader supply chain.

However, the hLR analysis suggests that these trade-offs can be mitigated. Under SSP2, the required learning rates to offset increases in material resource use and ozone depletion (4.57% and 3.36%) are below typical literature values (10–25%), indicating that simultaneous environmental improvements are achievable. Therefore, burden shifting should be interpreted as a transitional effect of system transformation rather than a fundamental limitation, provided sufficient technological learning is achieved .

Role of Environmental Learning: Environmental learning theory provides a key framework for assessing future environmental improvements, as it links reductions in impacts to accumulated production and technological experience. In this study, a quasi S-curve model is applied to capture the non-linear evolution of PLA technology, reflecting an initial slow phase, followed by rapid improvement, and eventual saturation. Compared to models assuming constant improvement, this approach better represents diminishing returns as technologies mature.

However, the effectiveness of learning is constrained by both technological capability and system-level conditions. While higher environmental learning rates (eLR) accelerate reductions in impacts, the overall improvement remains strongly influenced by background system decarbonization. In addition, technological motivation, represented by the doubling time (Td), plays a critical role in determining how quickly learning effects materialize. Faster growth (shorter Td) amplifies cumulative learning and leads to significantly lower impacts, whereas slower expansion limits environmental progress even under the same learning rate.

Combined Effects (Learning, Scaling and Motivation): Environmental improvements are driven by the combined effects of learning, scaling, and technological motivation. While the environmental learning rate (eLR) determines the reduction in impacts per unit, scaling—reflected by cumulative production—controls how quickly these improvements materialize. Results show that, at a fixed eLR of 5%, impacts decrease to 0.26 kg CO_2-eq/FU by 2050 with a short doubling time (3.56 years), compared to 0.59 kg CO_2-eq/FU under slower growth (10 years). This demonstrates that technological capability alone is insufficient, and strong growth dynamics are essential to achieve substantial environmental progress.

Methodological Limitations and Uncertainty: The results of this study are subject to several methodological limitations and uncertainties. First, the prospective background system, based on IAM outputs (REMIND) and SSP scenarios, relies on highly aggregated representations that may not accurately capture PLA-specific production pathways. In addition, the use of EXIOBASE with a 2019 baseline and aggregated sector definitions, as well as the application of global (GLO) datasets from ecoinvent where regional data are unavailable, may introduce biases and limit the representativeness of the results.

At the model level, the quasi S-curve learning approach assumes simplified technological evolution, while the

eLR and hLR are derived from literature or theoretical estimates rather than PLA-specific empirical data. Furthermore, SSP scenarios introduce additional uncertainty due to differing assumptions on future energy systems, policy developments, and technological progress. As background system changes dominate the results, the projected reductions in environmental impacts may be partially overestimated, while the contribution of foreground technological improvements may be underestimated. Therefore, the findings should be interpreted as indicative trends and relative comparisons rather than precise predictions of future environmental performance.

Future Research Directions: Future research should improve the robustness of prospective LCA for emerging materials such as PLA by developing PLA-specific inventory data and refining learning parameters based on empirical evidence. Integrating hybrid LCA approaches could enhance supply chain completeness and reduce truncation errors. In addition, exploring a wider range of IAM scenarios and multi-impact optimization methods would help address burden shifting. Finally, linking prospective LCA with absolute sustainability frameworks could provide more meaningful long-term benchmarks.

Stakeholder Recommendations: Policymakers should support renewable-energy deployment and low-carbon industrial infrastructure through stronger climate policies and incentives. Industrial stakeholders should improve process efficiency and reduce dependence on fossil-based transport and energy systems. In addition, researchers should further develop PLA-specific learning-rate datasets and improve prospective LCA methodologies to reduce uncertainties in future assessments. Coordinated actions across policy, industry, and supply chains are essential to avoid burden shifting and achieve long-term sustainability goals.

5. Conclusion

This study demonstrates that the environmental performance of PLA production is strongly influenced by both foreground technological learning and background system decarbonization. Under SSP1 and SSP2, climate change impacts decrease substantially over time, while burden-shifting risks toward upstream transport and supply chains remain important considerations. The results show that historical learning effects combined with rapid market growth can enable climate neutrality and mitigate burden-shifting impacts by 2050. However, achieving long-term sustainability requires not only efficiency improvements in the foreground system, but also broader transformations in energy systems, transport, and global supply chains. Overall, the study highlights the importance of integrating prospective LCA, EEIO analysis, and environmental learning approaches to support future sustainable development of bio-based plastics.

6. References

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[5] https://plasticseurope.org

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[7] https://ec.europa.eu/eurostat

7. Joker Question

Anqi Yu：

Your conclusion that PLA can achieve climate neutrality by 2050 relies on a proxy learning rate of 15% and a short-term growth rate of 21.5% (doubling time 3.56 years). How would your conclusion change if you instead used (a) a learning rate of 5% or (b) a growth rate of 10% (doubling time ~10 years)? Which of these two assumptions introduces greater uncertainty, and what additional analysis would you need to defend your recommendation in a real Danish policy context?

Yining Gan:

The results suggest that achieving carbon neutrality for PLA by 2050 relies on the combined effects of background system decarbonization (SSP scenarios) and foreground technological learning. However, given the high uncertainty in SSP-based projections, how robust is your conclusion? Specifically, if background decarbonization is slower than expected, could the foreground system alone (through learning effects) still achieve the environmental target?

Jiayue Che:

Which methodological limitation introduces the greatest uncertainty in this study, and how could future research improve the reliability of the prospective LCA results?

8. Feedback Fruit Reflections

Based on the Feedback Fruit sessions, the KP report was improved by clarifying the prospective LCA methodology, including the integration of premise, SSP scenarios, and environmental learning curves. The foreground system definition and cradle-to-gate boundary were also explained more clearly. In addition, the discussion was strengthened by adding clearer explanations of trade-offs, burden shifting, and methodological limitations. Finally, the structure, figure descriptions, and interpretation of results were revised to improve clarity and consistency throughout the report.

9. Individual Contributions

		Anqi Yu	**Yining Gan**	**Jiayue Che**
1. Introduction	1.1 Project Overview and Case Specifics	×		
	1.2 Strategic Importance of Prospective Modeling for PLA	×	×	
	1.3 Macro-sector Definition and Mapping	×	×	
2. Method	2.1 Life Cycle Assessment (LCA) Approach		×	
	2.2 Prospective Approach		×	×
	2.3 Life Cycle Inventory (LCI)		×	×
	2.4 Life Cycle Impact Assessment (LCIA)		×	×
3. Results	3.1 Life Cycle Impact Assessment Results		×	×

	3.2 Background System Analysis		×	×
	3.3 Environmental Learning Effects and Motivational Factors		×	×
	3.4 Suggestions for a Sustainable Roadmap			×
4. Discussion			×	×
5. Conclusion		×		

Appendices

Appendix I: Results and Data

1. TOP 5 CONTRIBUTORS

('DK', 'Sea and coastal water transportation services'): 0.056 kt CO2e pr MEUR

('DK', 'Plastics, basic'): 0.052 kt CO2e pr MEUR

('DK', 'Steam and hot water supply services'): 0.019 kt CO2e pr MEUR

('WA', 'Sea and coastal water transportation services'): 0.009 kt CO2e pr MEUR

('WM', 'Sea and coastal water transportation services'): 0.007 kt CO2e pr MEUR

	# Step 1	# Step 2	# Step 3	# Step 4
	缺少: 0 (0%) 非重复: 2 (14%)	缺少: 0 (0%) 非重复: 6 (43%)	缺少: 0 (0%) 非重复: 6 (43%)	缺少: 0 (0%) 非重复: 6 (43%)
	最小值 ... 最大值 0.052046556695599...	最小值 0.0 最大值 0.02910930751271364	最小值 0.0 最大值 0.01528723404259725	最小值 ... 最大值 0.0071641400912486...
DK - Plastics,	0.052046556695599995	0.0	0.0	0.0
AT - Paddy ric	0.0	0.0	0.0	0.0
KR - Electricity	0.0	0.0	0.0	0.0
KR - Electricity	0.0	0.0	0.0	0.0
KR - Electricity	0.0	0.0	0.0	0.0
DK - Sea and	0.0	0.02910930751271364	0.01528723404259725	0.007164140091248696
DK - Steam ar	0.0	0.016331855746546192	0.0017637198894769899	0.0
US - Crude pe	0.0	0.003032691310678432	0.0	0.0
NO - Crude p	0.0	0.0026571113032554804	0.0	0.0
DK - Pigs	0.0	0.002096826786498459	0.0	0.0
WM - Sea and	0.0	0.0	0.003303799829934246	0.0021963424431345135
WA - Sea and	0.0	0.0	0.0032060357405602536	0.0026090776412143473
DE - Sea and	0.0	0.0	0.0018733261922941166	0.0018075057666368682
WM - Suppor	0.0	0.0	0.0	0.0009196580290510304

2. Projected climate change impacts, ozone depletion and material resources under the SSP1 and SSP2 scenarios

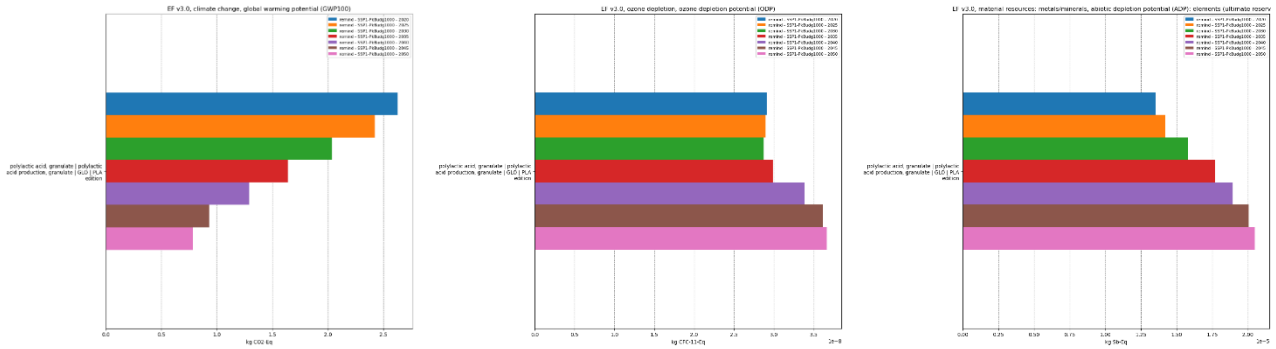


Figure 1. Projected climate change impacts, ozone depletion and material resources under the SSP1 scenario (2020–2050).

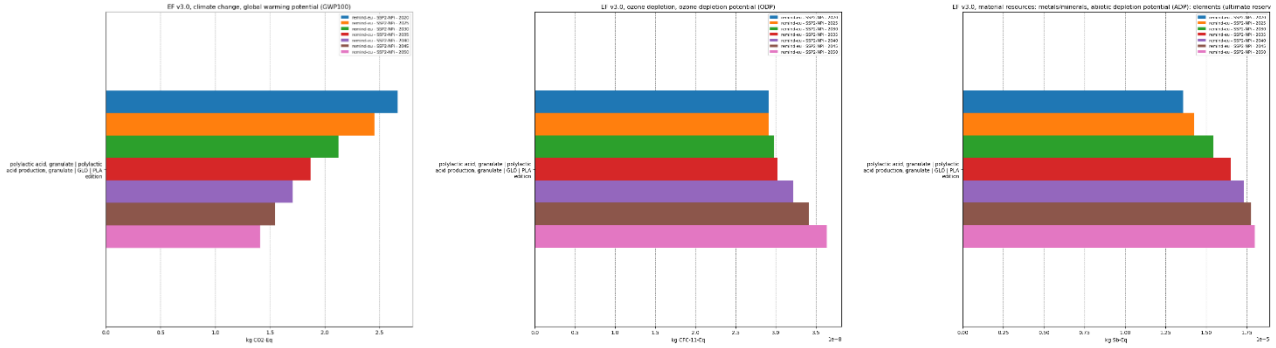


Figure 2. Projected climate change impacts, ozone depletion and material resources under the SSP2 scenario (2020–2050).

3. Comparison of emission contributions for the Danish low-voltage electricity market under SSP1 and SSP2 (2020 vs 2050)

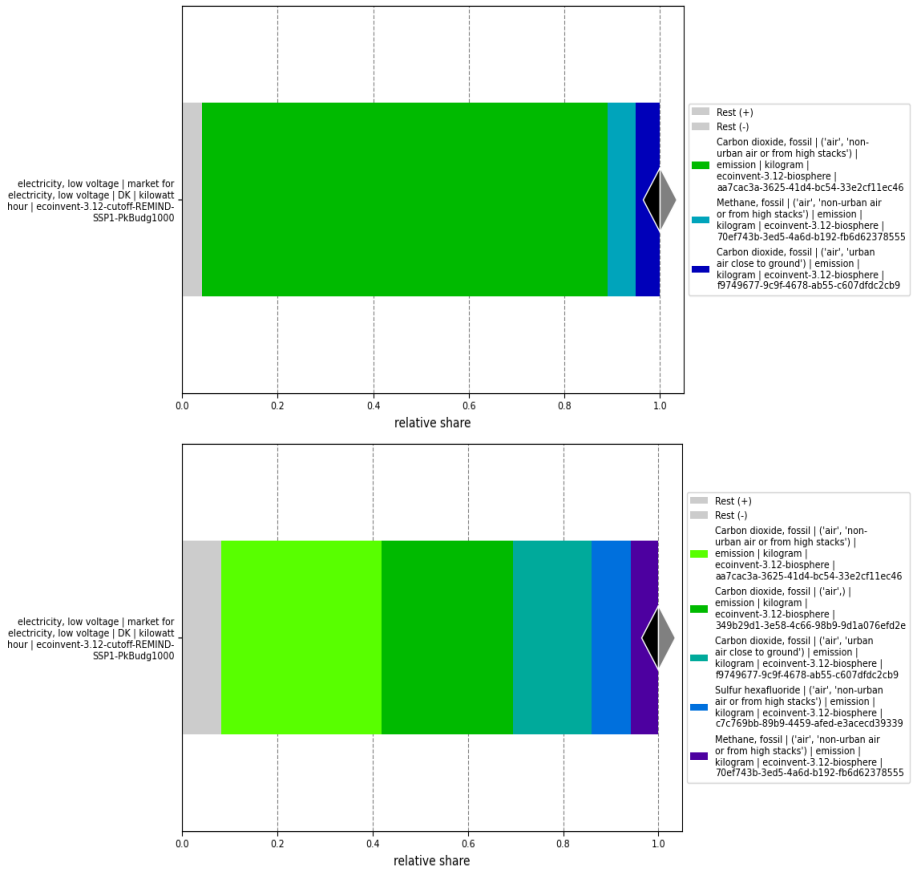


Figure 1. Relative contribution of elementary emissions to the environmental impacts of the Danish low-voltage electricity market

under SSP1 in 2020 and 2050.

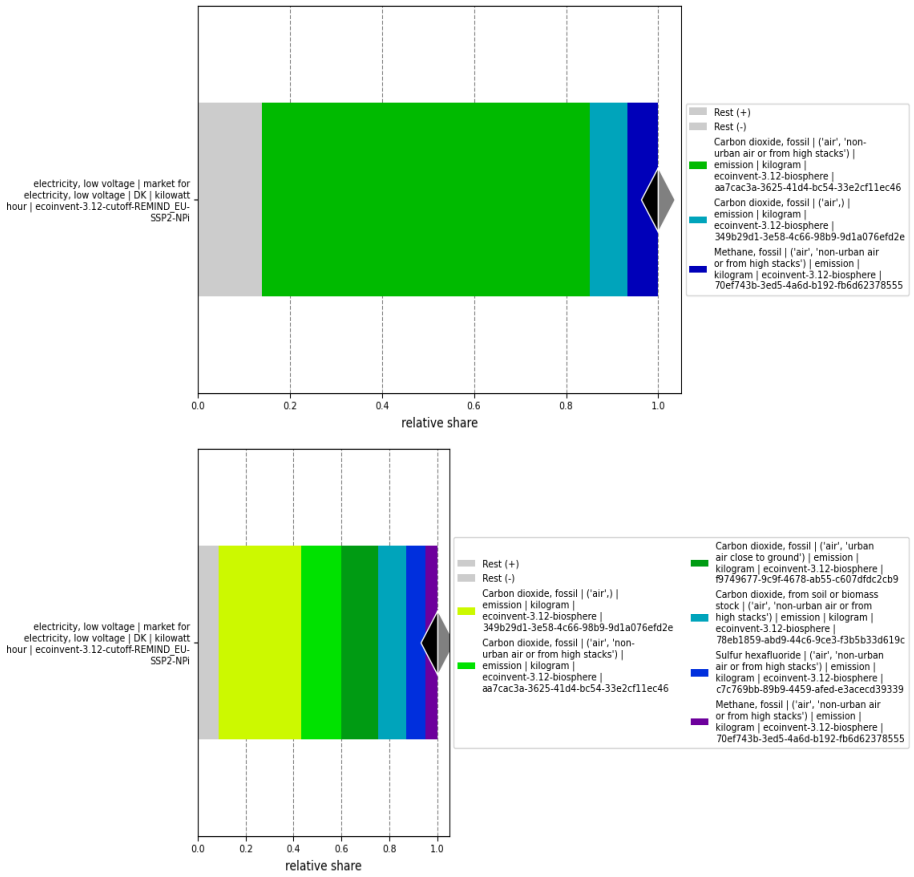


Figure 2. Relative contribution of elementary emissions to the environmental impacts of the Danish low-voltage electricity market under SSP1 in 2020 and 2050.

4. AB simulation data

(1) SSP1

Table 1. climate impact in AB simulation results for SSP1

EF v3.0, climate change, global warming potential (GWP100)						
remind - SSP1-PkBudg1000 - 2020	remind - SSP1-PkBudg1000 - 2025	remind - SSP1-PkBudg1000 - 2030	remind - SSP1-PkBudg1000 - 2035	remind - SSP1-PkBudg1000 - 2040	remind - SSP1-PkBudg1000 - 2045	remind - SSP1-PkBudg1000 - 2050
2.628811313	2.421270563	2.036650102	1.641108614	1.289757217	0.930561095	0.786236145

Table 2. Material impact in AB simulation results for SSP1

EF v3.0, material resources: metals/minerals, abiotic depletion potential (ADP): elements (ultimate reserves)						
remind - SSP1-PkBudg1000 - 2020	remind - SSP1-PkBudg1000 - 2025	remind - SSP1-PkBudg1000 - 2030	remind - SSP1-PkBudg1000 - 2035	remind - SSP1-PkBudg1000 - 2040	remind - SSP1-PkBudg1000 - 2045	remind - SSP1-PkBudg1000 - 2050
1.35577E-05	1.42078E-05	1.58149E-05	1.76819E-05	1.89433E-05	2.00678E-05	2.04986E-05

Table 3. Ozone depletion in AB simulation results for SSP1

EF v3.0, ozone depletion, ozone depletion potential (ODP)						
remind - SSP1-PkBudg1000 - 2020	remind - SSP1-PkBudg1000 - 2025	remind - SSP1-PkBudg1000 - 2030	remind - SSP1-PkBudg1000 - 2035	remind - SSP1-PkBudg1000 - 2040	remind - SSP1-PkBudg1000 - 2045	remind - SSP1-PkBudg1000 - 2050
2.90801E-08	2.89875E-08	2.87582E-08	2.98771E-08	3.38472E-08	3.61276E-08	3.66581E-08

(2) SSP2

Table 4. climate impact in AB simulation results for SSP2

EF v3.0, climate change, global warming potential (GWP100)						
remind-eu - SSP2-NPi - 2020	remind-eu - SSP2-NPi - 2025	remind-eu - SSP2-NPi - 2030	remind-eu - SSP2-NPi - 2035	remind-eu - SSP2-NPi - 2040	remind-eu - SSP2-NPi - 2045	remind-eu - SSP2-NPi - 2050
2.667007673	2.453263392	2.122324219	1.870826551	1.706921152	1.545521268	1.408054483

Table 5. Material impact in AB simulation results for SSP2

EF v3.0, material resources: metals/minerals, abiotic depletion potential (ADP): elements (ultimate reserves)						
remind-eu - SSP2-NPi - 2020	remind-eu - SSP2-NPi - 2025	remind-eu - SSP2-NPi - 2030	remind-eu - SSP2-NPi - 2035	remind-eu - SSP2-NPi - 2040	remind-eu - SSP2-NPi - 2045	remind-eu - SSP2-NPi - 2050
1.35806E-05	1.42621E-05	1.54415E-05	1.65408E-05	1.7307E-05	1.77731E-05	1.80029E-05

Table 6. Ozone depletion in AB simulation results for SSP2

EF v3.0, ozone depletion, ozone depletion potential (ODP)						
remind-eu - SSP2-NPi - 2020	remind-eu - SSP2-NPi - 2025	remind-eu - SSP2-NPi - 2030	remind-eu - SSP2-NPi - 2035	remind-eu - SSP2-NPi - 2040	remind-eu - SSP2-NPi - 2045	remind-eu - SSP2-NPi - 2050
2.9091E-08	2.91409E-08	2.98226E-08	3.02076E-08	3.21366E-08	3.41542E-08	3.63625E-08

5. EI value

T_d= 3.56 year:

	Td (years)	Scenario	eLR (%)	Year	Background EI	Foreground EI (S-curve)	Comprehensive EI
2	3.56	SSP1 (PkBudg1000)	3.3	2020	2.628811313	2.628811313	2.628811313
3	3.56	SSP1 (PkBudg1000)	3.3	2025	2.421270563	2.567843945	2.360303195
4	3.56	SSP1 (PkBudg1000)	3.3	2030	2.036650102	2.502971862	1.910810651
5	3.56	SSP1 (PkBudg1000)	3.3	2035	1.641108614	2.438247115	1.450544416
6	3.56	SSP1 (PkBudg1000)	3.3	2040	1.289757217	2.375409862	1.036355766
7	3.56	SSP1 (PkBudg1000)	3.3	2045	0.930561095	2.315061688	0.61681147
8	3.56	SSP1 (PkBudg1000)	3.3	2050	0.786236145	2.257344837	0.414769669
9	3.56	SSP1 (PkBudg1000)	5	2020	2.628811313	2.628811313	2.628811313
10	3.56	SSP1 (PkBudg1000)	5	2025	2.421270563	2.536831995	2.329291245
11	3.56	SSP1 (PkBudg1000)	5	2030	2.036650102	2.441098126	1.848936915
12	3.56	SSP1 (PkBudg1000)	5	2035	1.641108614	2.347821891	1.360119192
13	3.56	SSP1 (PkBudg1000)	5	2040	1.289757217	2.259452877	0.920398781
14	3.56	SSP1 (PkBudg1000)	5	2045	0.930561095	2.176657648	0.47840743
15	3.56	SSP1 (PkBudg1000)	5	2050	0.786236145	2.099415486	0.256840318
16	3.56	SSP1 (PkBudg1000)	10	2020	2.628811313	2.628811313	2.628811313
17	3.56	SSP1 (PkBudg1000)	10	2025	2.421270563	2.44724358	2.23970283
18	3.56	SSP1 (PkBudg1000)	10	2030	2.036650102	2.270551014	1.678389803
19	3.56	SSP1 (PkBudg1000)	10	2035	1.641108614	2.110405558	1.122702859
20	3.56	SSP1 (PkBudg1000)	10	2040	1.289757217	1.96958012	0.630526024
21	3.56	SSP1 (PkBudg1000)	10	2045	0.930561095	1.847218893	0.148968675
22	3.56	SSP1 (PkBudg1000)	10	2050	0.786236145	1.741390131	-0.101185037
23	3.56	SSP1 (PkBudg1000)	15	2020	2.628811313	2.628811313	2.628811313
24	3.56	SSP1 (PkBudg1000)	15	2025	2.421270563	2.360202	2.15266125
25	3.56	SSP1 (PkBudg1000)	15	2030	2.036650102	2.116716777	1.524555566
26	3.56	SSP1 (PkBudg1000)	15	2035	1.641108614	1.912269334	0.924566635
27	3.56	SSP1 (PkBudg1000)	15	2040	1.289757217	1.746103143	0.407049047
28	3.56	SSP1 (PkBudg1000)	15	2045	0.930561095	1.612783656	-0.085466562
29	3.56	SSP1 (PkBudg1000)	15	2050	0.786236145	1.506348518	-0.33622665

30	3.56	SSP2 (NPi)	3.3	2020	2.667007673	2.667007673	2.667007673
31	3.56	SSP2 (NPi)	3.3	2025	2.453263392	2.605154455	2.391410174
32	3.56	SSP2 (NPi)	3.3	2030	2.122324219	2.539339788	1.994656334
33	3.56	SSP2 (NPi)	3.3	2035	1.870826551	2.473674597	1.677493475
34	3.56	SSP2 (NPi)	3.3	2040	1.706921152	2.409924324	1.449837803
35	3.56	SSP2 (NPi)	3.3	2045	1.545521268	2.348699298	1.227212893
36	3.56	SSP2 (NPi)	3.3	2050	1.408054483	2.290143827	1.031190637
37	3.56	SSP2 (NPi)	5	2020	2.667007673	2.667007673	2.667007673
38	3.56	SSP2 (NPi)	5	2025	2.453263392	2.573691905	2.359947624
39	3.56	SSP2 (NPi)	5	2030	2.122324219	2.476567033	1.931883579
40	3.56	SSP2 (NPi)	5	2035	1.870826551	2.381935504	1.585754382
41	3.56	SSP2 (NPi)	5	2040	1.706921152	2.292282497	1.332195976
42	3.56	SSP2 (NPi)	5	2045	1.545521268	2.208284262	1.086797857
43	3.56	SSP2 (NPi)	5	2050	1.408054483	2.129919778	0.870966588
44	3.56	SSP2 (NPi)	10	2020	2.667007673	2.667007673	2.667007673
45	3.56	SSP2 (NPi)	10	2025	2.453263392	2.48280178	2.269057499
46	3.56	SSP2 (NPi)	10	2030	2.122324219	2.303541888	1.758858434
47	3.56	SSP2 (NPi)	10	2035	1.870826551	2.141069535	1.344888413
48	3.56	SSP2 (NPi)	10	2040	1.706921152	1.998197918	1.038111397
49	3.56	SSP2 (NPi)	10	2045	1.545521268	1.874058795	0.75257239
50	3.56	SSP2 (NPi)	10	2050	1.408054483	1.766692352	0.507739162
51	3.56	SSP2 (NPi)	15	2020	2.667007673	2.667007673	2.667007673
52	3.56	SSP2 (NPi)	15	2025	2.453263392	2.394495494	2.180751213
53	3.56	SSP2 (NPi)	15	2030	2.122324219	2.147472455	1.602789001
54	3.56	SSP2 (NPi)	15	2035	1.870826551	1.940054412	1.14387329
55	3.56	SSP2 (NPi)	15	2040	1.706921152	1.771473843	0.811387322
56	3.56	SSP2 (NPi)	15	2045	1.545521268	1.636217237	0.514730832
57	3.56	SSP2 (NPi)	15	2050	1.408054483	1.528235608	0.269282418

T_d= 10 year:

1	Td (years)	Scenario	eLR (%)	Year	Background EI	Foreground EI (S-curve)	Comprehensive EI
2	10	SSP1 (PkBudg1000)	3.3	2020	2.628811313	2.628811313	2.628811313
3	10	SSP1 (PkBudg1000)	3.3	2025	2.421270563	2.607938408	2.400397658
4	10	SSP1 (PkBudg1000)	3.3	2030	2.036650102	2.586059324	1.993898113
5	10	SSP1 (PkBudg1000)	3.3	2035	1.641108614	2.563494017	1.575791318
6	10	SSP1 (PkBudg1000)	3.3	2040	1.289757217	2.540511892	1.201457796
7	10	SSP1 (PkBudg1000)	3.3	2045	0.930561095	2.517327354	0.819077136
8	10	SSP1 (PkBudg1000)	3.3	2050	0.786236145	2.494103914	0.651528746
9	10	SSP1 (PkBudg1000)	5	2020	2.628811313	2.628811313	2.628811313
10	10	SSP1 (PkBudg1000)	5	2025	2.421270563	2.597104461	2.389563711
11	10	SSP1 (PkBudg1000)	5	2030	2.036650102	2.564110858	1.971949647
12	10	SSP1 (PkBudg1000)	5	2035	1.641108614	2.530343296	1.542640597
13	10	SSP1 (PkBudg1000)	5	2040	1.289757217	2.496226183	1.157172087
14	10	SSP1 (PkBudg1000)	5	2045	0.930561095	2.462090985	0.763840767
15	10	SSP1 (PkBudg1000)	5	2050	0.786236145	2.428184981	0.585609813
16	10	SSP1 (PkBudg1000)	10	2020	2.628811313	2.628811313	2.628811313
17	10	SSP1 (PkBudg1000)	10	2025	2.421270563	2.564912711	2.357371961
18	10	SSP1 (PkBudg1000)	10	2030	2.036650102	2.49988369	1.907722479
19	10	SSP1 (PkBudg1000)	10	2035	1.641108614	2.434872173	1.447169474
20	10	SSP1 (PkBudg1000)	10	2040	1.289757217	2.370770888	1.031716792
21	10	SSP1 (PkBudg1000)	10	2045	0.930561095	2.308226234	0.609976016
22	10	SSP1 (PkBudg1000)	10	2050	0.786236145	2.247674517	0.405099349
23	10	SSP1 (PkBudg1000)	15	2020	2.628811313	2.628811313	2.628811313
24	10	SSP1 (PkBudg1000)	15	2025	2.421270563	2.532220572	2.324679822
25	10	SSP1 (PkBudg1000)	15	2030	2.036650102	2.436205226	1.844044015
26	10	SSP1 (PkBudg1000)	15	2035	1.641108614	2.342566307	1.354863608
27	10	SSP1 (PkBudg1000)	15	2040	1.289757217	2.252589112	0.913535016
28	10	SSP1 (PkBudg1000)	15	2045	0.930561095	2.16709684	0.468846622
29	10	SSP1 (PkBudg1000)	15	2050	0.786236145	2.086542429	0.243967261

30	10	SSP2 (NPi)	3.3	2020	2.667007673	2.667007673	2.667007673
31	10	SSP2 (NPi)	3.3	2025	2.453263392	2.645831487	2.432087206
32	10	SSP2 (NPi)	3.3	2030	2.122324219	2.623634502	2.078951048
33	10	SSP2 (NPi)	3.3	2035	1.870826551	2.600741323	1.804560201
34	10	SSP2 (NPi)	3.3	2040	1.706921152	2.577425271	1.61733875
35	10	SSP2 (NPi)	3.3	2045	1.545521268	2.553903864	1.432417459
36	10	SSP2 (NPi)	3.3	2050	1.408054483	2.530342989	1.271389799
37	10	SSP2 (NPi)	5	2020	2.667007673	2.667007673	2.667007673
38	10	SSP2 (NPi)	5	2025	2.453263392	2.634840123	2.421095842
39	10	SSP2 (NPi)	5	2030	2.122324219	2.601367127	2.056683673
40	10	SSP2 (NPi)	5	2035	1.870826551	2.567108926	1.770927804
41	10	SSP2 (NPi)	5	2040	1.706921152	2.532496095	1.572409574
42	10	SSP2 (NPi)	5	2045	1.545521268	2.497864916	1.376378511
43	10	SSP2 (NPi)	5	2050	1.408054483	2.46346626	1.20451307
44	10	SSP2 (NPi)	10	2020	2.667007673	2.667007673	2.667007673
45	10	SSP2 (NPi)	10	2025	2.453263392	2.602180631	2.38843635
46	10	SSP2 (NPi)	10	2030	2.122324219	2.536206744	1.99152329
47	10	SSP2 (NPi)	10	2035	1.870826551	2.470250617	1.674069495
48	10	SSP2 (NPi)	10	2040	1.706921152	2.405217947	1.445131426
49	10	SSP2 (NPi)	10	2045	1.545521268	2.341764526	1.220278121
50	10	SSP2 (NPi)	10	2050	1.408054483	2.280332998	1.021379808
51	10	SSP2 (NPi)	15	2020	2.667007673	2.667007673	2.667007673
52	10	SSP2 (NPi)	15	2025	2.453263392	2.569013478	2.355269197
53	10	SSP2 (NPi)	15	2030	2.122324219	2.471603039	1.926919585
54	10	SSP2 (NPi)	15	2035	1.870826551	2.376603556	1.580422434
55	10	SSP2 (NPi)	15	2040	1.706921152	2.285319002	1.325232481
56	10	SSP2 (NPi)	15	2045	1.545521268	2.198584536	1.077098131
57	10	SSP2 (NPi)	15	2050	1.408054483	2.116859677	0.857906487

6. Calculation procedure of the hLR

using SSP2 (2020–2050) with climate change selected as the environmental impact (EI) indicator; two technology doubling times were applied for the combination analysis.

1. T_d= 3.56 y

Task 4a

In this example, historical or explorative environmental learning rates of a specific technology are not available/unknown. The LCA practitioner/sustainability engineer would like to pre-screen the potential of an emerging or incumbent technology to reach a specific environmental goal by assuming that learning effects will affect the environmental impacts within the foreground system over the prospective time frame. Therefore, the hypothetical learning rate (i.e., the learning rate required to achieve a specific future goal) is calculated and compared to generic historical values to assess feasibilty.

Variable	Value	Unit	Comment
t1	2020	a	Starting point of the prospective time frame
t2	2050	a	End point of the prospective time frame
$EI_{goal,t2}$	0	kg CO_2-eq	Environmental goal in t2. In this example: carbon emission neutrality (= 0 kg CO_2-eq climate change impact)
$EI_{background,t1}$	2.667007673	kg CO_2-eq	Activity Browser simulation result in t1
$EI_{background,t2}$	1.408054483	kg CO_2-eq	Activity Browser simulation result in t2
τ_i	30	a	Prospective time interval
r	21.5	%	Yearly growth rate
T_d	3.56	a	Technology doubling time
α_e	0.128515684	-	Power constant
hLR_p	8.52	%	Hypothetical learning rate (if $\alpha_{e,h} > 0$)
hLR_n	-8.52	%	Hypothetical learning rate (if $\alpha_{e,h} < 0$)

$$EI_{comprehensive} = EI_{background,k} - EI_{progress,foreground,i}$$

$$EI_{goal,t2} = EI_{background,t2} - EI_{background,t1} + EI_{background,t1} \cdot \left(2^{\frac{\tau_i}{T_d}}\right)^{-\alpha_{e,h}}$$

$$\alpha_{e,h} = \frac{ln\left(\frac{\left|EI_{goal,t2} - EI_{background,t2} + EI_{background,t1}\right|}{\left|EI_{background,t1}\right|}\right)}{-\left(\frac{\tau_i}{T_d}\right) \cdot ln2}$$

$$hLR(\%) = (1 - 2^{-\alpha_{e,h}}) \cdot 100\%, if\ \alpha_{e,h} > 0$$

$$hLR(\%) = (2^{-\alpha_{e,h}} - 1) \cdot 100\%, if\ \alpha_{e,h} < 0$$

Burden shifting:

① material resources:

Task 4b			
In this example, burden shifting is detected in the mineral resource use impact category (i.e., even though introducing a novel technology decreases the climate change impact, it results in increased mineral resource use). The impacts are increased from 0.8 kg deprieved/FU to 0.9 kg deprieved/FU between 2025 and 2050. If we assume that technological learning and environmental progress will happen in the foreground system over the prospective time interval, could it be enough to neutralize the burden shifting observed in the background system? What hypothetical learning rate would be required?			
Variable	**Value**	**Unit**	**Comment**
t1	2020	a	Starting point of the prospective time frame
t2	2050	a	End point of the prospective time frame
$EI_{goal,t2}$	1.35806E-05	kg deprieved	Environmental goal in t2. In this example: to neutralize burden shifting. Thus, the environmental goal equals the environmental impacts at the starting point in t1 (2025).
$EI_{background,t1}$	1.35806E-05	kg deprieved	Activity Browser simulation result in t1
$EI_{background,t2}$	1.80029E-05	kg deprieved	Activity Browser simulation result in t2
τ_i	30	a	Prospective time interval
r	21.5	%	Yearly growth rate
T_d	3.56	a	Technology doubling time
α_e	0.067450828	-	Power constant
hLR_p	**4.57**	**%**	**Hypothetical learning rate (if $\alpha_{e,h} > 0$)**
hLR_n	**-4.57**	**%**	**Hypothetical learning rate (if $\alpha_{e,h} < 0$)**

$$EI_{goal,t2} = EI_{background,t1}$$

② ozone depletion:

Task 4b			
In this example, burden shifting is detected in the mineral resource use impact category (i.e., even though introducing a novel technology decreases the climate change impact, it results in increased mineral resource use). The impacts are increased from 0.8 kg deprieved/FU to 0.9 kg deprieved/FU between 2025 and 2050. If we assume that technological learning and environmental progress will happen in the foreground system over the prospective time interval, could it be enough to neutralize the burden shifting observed in the background system? What hypothetical learning rate would be required?			
Variable	**Value**	**Unit**	**Comment**
t1	2020	a	Starting point of the prospective time frame
t2	2050	a	End point of the prospective time frame
$EI_{goal,t2}$	2.91E-08	kg deprieved	Environmental goal in t2. In this example: to neutralize burden shifting. Thus, the environmental goal equals the environmental impacts at the starting point in t1 (2025).
$EI_{background,t1}$	2.91E-08	kg deprieved	Activity Browser simulation result in t1
$EI_{background,t2}$	3.64E-08	kg deprieved	Activity Browser simulation result in t2
τ_i	30	a	Prospective time interval
r	21.5	%	Yearly growth rate
T_d	3.56	a	Technology doubling time
α_e	0.049241146	-	Power constant
hLR_p	**3.36**	**%**	**Hypothetical learning rate (if $\alpha_{e,h} > 0$)**
hLR_n	**-3.36**	**%**	**Hypothetical learning rate (if $\alpha_{e,h} < 0$)**

$$EI_{goal,t2} = EI_{background,t1}$$

2. T_d= 10 y

Task 4a			
In this example, historical or explorative environmental learning rates of a specific technology are not available/unknown. The LCA practitioner/sustainability engineer would like to pre-screen the potential of an emerging or incumbent technology to reach a specific environmental goal by assuming that learning effects will affect the environmental impacts within the foreground system over the prospective time frame. Therefore, the hypothetical learning rate (i.e., the learning rate required to achieve a specific future goal) is calculated and compared to generic historical values to assess feasibilty.			
Variable	**Value**	**Unit**	**Comment**
t1	2020	a	Starting point of the prospective time frame
t2	2050	a	End point of the prospective time frame
$EI_{goal,t2}$	0	kg CO_2-eq	Environmental goal in t2. In this example: carbon emission neutrality (= 0 kg CO_2-eq climate change impact)
$EI_{background,t1}$	2.667007673	kg CO_2-eq	Activity Browser simulation result in t1
$EI_{background,t2}$	1.408054483	kg CO_2-eq	Activity Browser simulation result in t2
τ_i	30	a	Prospective time interval
r	7.18	%	Yearly growth rate
T_d	10.00	a	Technology doubling time
α_e	0.360999111	-	Power constant
hLR_p	22.14	%	Hypothetical learning rate (if $\alpha_{e,h} > 0$)
hLR_n	-22.14	%	Hypothetical learning rate (if $\alpha_{e,h} < 0$)

$$EI_{comprehensive} = EI_{background,k} - EI_{progress,foreground,i}$$

$$EI_{goal,t2} = EI_{background,t2} - EI_{background,t1} + EI_{background,t1} \cdot \left(2^{\frac{\tau_i}{T_d}}\right)^{-\alpha_{e,h}}$$

$$\alpha_{e,h} = \frac{ln\left(\frac{\left|EI_{goal,t2} - EI_{background,t2} + EI_{background,t1}\right|}{\left|EI_{background,t1}\right|}\right)}{-\left(\frac{\tau_i}{T_d}\right) \cdot ln2}$$

$$hLR(\%) = (1 - 2^{-\alpha_{e,h}}) \cdot 100\%, if\ \alpha_{e,h} > 0$$

$$hLR(\%) = (2^{-\alpha_{e,h}} - 1) \cdot 100\%, if\ \alpha_{e,h} < 0$$

7. Suggestions for a sustainable roadmap

Table 1. AESA comparison results for the Danish PLA system.

AESA comparison	
soSOS	2.54E-02
Case Study	1.97485207872227
Level of Exceedance	77.6

Table 2. Calculated eLR for the climate change target under SSP1.

Prospective LCA	Climate change as kg CO_2
Growth rate [%]	15.0%
Doubling time [year]	4.96
SSP1-PkBudg1000-Systems elasticity (a)	1.63E-01
SSP1-PkBudg1000-eLR for the target [%]	10.7%

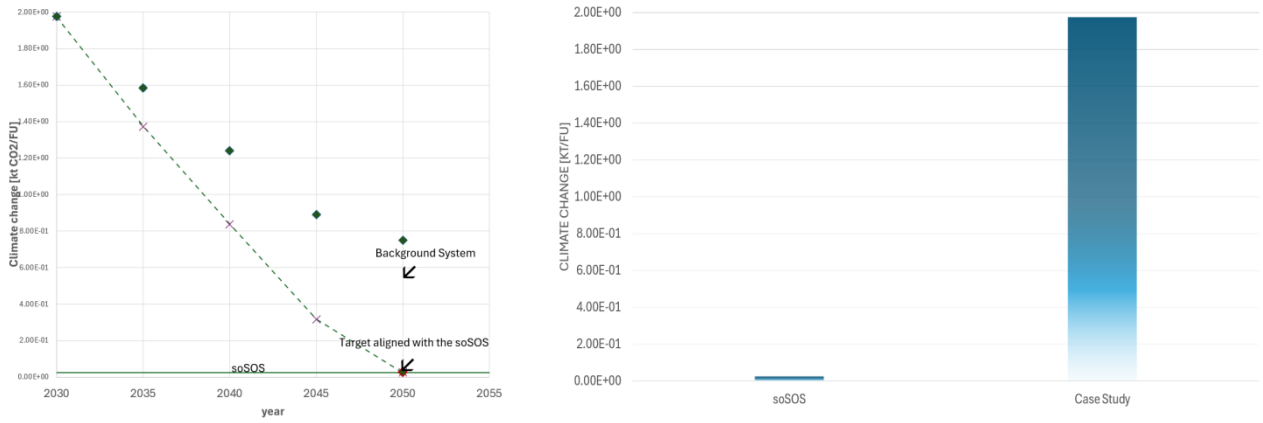


Figure 1. Comparison between soSOS and projected climate impacts of PLA production.

Appendix II: Equation

1. growth rate(r), doubling time(T_d), the number of doublings(n_d)

S_i = 2.424 million tonnes, S_0 = 0.916 million tonnes, Δt = 5 years.

Based on these values, the yearly growth rate (r) was calculated using the following equation:

$$r(\%) = (e^{\frac{\ln(\frac{S_i}{S_0})}{\Delta t}} - 1) \cdot 100\%$$

where:

S_i = cumulative capacity at time i
S_0 = initial capacity
Δt = time difference between the initial capacity and capacity at time i

The calculated growth rate was approximately 21.5% per year. Based on this growth rate, the technology doubling time (T_d) was estimated using:

$$T_d(a) = t\frac{\ln 2}{\ln\left(1+\frac{r}{100}\right)}$$

where:

r = growth rate specified for time t (1 year)

The resulting technology doubling time was approximately 3.56 years. The number of doublings (n_d) over the prospective assessment period was then calculated as:

$$n_d = \frac{\Delta t}{T_d}$$

where:

Δt = time interval for your KP (between 2025 and 2050)

2. Hypothetical Learning Rate (hLR)

The comprehensive prospective environmental impact was calculated by combining background system development with hypothetical technological learning in the foreground system:

$$EI_{comprehensive} = EI_{background,t_2} - EI_{background,t_1} + EI_{background,t_1} \cdot 2^{-\alpha_{e,h} \cdot \tau / T_d}$$

The hypothetical learning exponent was calculated as:

$$\alpha_{e,h} = \frac{\ln\left(\frac{EI_{goal,t_2} - EI_{background,t_2} + EI_{background,t_1}}{| EI_{background,t_1} |}\right)}{-\left(\frac{\tau}{T_d}\right)\ln 2}$$

and converted into the hypothetical learning rate:

$$hLR(\%) = (1 - 2^{-\alpha_{e,h}}) \times 100\%$$